Glossary

A

ABA design—A single-case design in which the response to the treatment condition is compared to baseline responses recorded before and after treatment

ABAB design—Extension to ABA design to include reintroduction of the treatment condition

Active consent—Verbally agreeing and signing a form consenting to participate in research

Active deception—Deliberately misleading research participants by giving them false information

Additive and interactive effects—Differences between groups are produced because of the combined effect of two or more threats to internal validity

Alpha level—The point at which one would reject the null hypothesis and accept the alternative hypothesis

Alternative hypothesis—The logical opposite of the null hypothesis

Analysis of covariance—Statistical test used when you have one quantitative DV and a mixture of categorical and quantitative IVs (the quantitative IV is called a *"covariate"*); a statistical procedure in which group means are compared after adjusting for pretest differences

Anchor—Descriptors placed on points on a rating scale

ANCOVA—Abbreviation for analysis of covariance

Animal rights—The belief that animals have rights similar to humans and should not be used in research

Animal welfare—Improving the laboratory conditions in which animals live and reducing the number of animals used in research

Anonymity—Identity of participants is unknown and therefore cannot be revealed

ANOVA—Abbreviation for analysis of variance.

Archived research data—Data (usually quantitative) originally used for a different research project and stored in a way that others can use it

Assent—Agreement from a minor to participate in research after receiving an age-appropriate explanation of the study

Assignment measure—Measure used to assign participants to experimental and control groups; those with scores below the cutoff score are assigned to one group, and those with scores above the cutoff are assigned to the other group

Attrition—Loss of participants because they don't show up or they drop out of the research study

Authority—A basis for acceptance of information because it is acquired from a highly respected source

Automation—The technique of totally automating the experimental procedures so that no researcher–participant interaction is required

Axial coding—Second stage of data analysis in GT; focus is on making concepts more abstract and ordering them into the theory

B

Bar graph—Graph that uses vertical bars to represent the data values of a categorical variable

Baseline—The target behavior of the participant in its naturally occurring state.

Beneficence—Acting for the benefit of others

Between-participants designs—Groups are produced by random assignment, and the different groups are exposed to the different levels of the independent variable (uses a between-participants IV)

Between-participants independent variable—Type of independent variable where different participants receive different levels of the independent variable

Biased sample—A nonrepresentative sample

Binary forced-choice approach—Participant must select from the two response choices provided with an item

Bradford Hill criteria for causation—Set of nine criteria used cumulatively to produce evidence of causation

C

Carryover effect—A sequencing effect that occurs when performance in a treatment condition is affected by the treatment condition that preceded it.

Case—A bounded system

Case study—Qualitative research method in which the researcher provides a detailed description and account of one or more cases

Categorical variable—Variable that varies by type or kind

Causal description—Description of the consequences of manipulating an independent variable

Causal explanation—Explaining the mechanisms through which a causal relationship operates

Causation validity—A synonym for internal validity

Cause—The factor that makes something else exist or change

Cause-and-effect relationship—Relationship in which changes in one variable produce changes in another variable

Ceiling effect—Situation where participants' pretest scores on the dependent variable are too high to allow for additional increases

Cell—Combination of levels of two or more independent variables

Cell mean—The average score of the participants in a single cell

Census—Collection of data from everyone in the population

Changing-criterion design—A single-case design in which a participant's behavior is gradually shaped by changing the criterion for success during successive treatment periods

Checklist—Participants asked to check all response categories that apply

Chi-square test for independence—Statistical test used to determine if a relationship observed in a contingency table is statistically significant

Clinical significance—Treatment effect is of sufficient magnitude to affect real-life functioning; a type of practical significance

Closed-ended question—A question where participants must select their answer from a set of predetermined response categories

Cluster—A collective type of unit that includes multiple elements

Cluster random sampling—Sampling method where clusters are randomly selected

Coefficient alpha—The most frequently used index of internal consistency

Cohen's *d*—The difference between two means in standard deviation units

Cohort-sequential design—Design that combines cross-sectional and longitudinal elements by following two or more age groups over time

Collective case study—Study of multiple cases for the purpose of comparison

Combination design—Single-case design used to identify the combined effects of multiple independent variables

Comparative case study—Another name for a collective case study

Compatibility thesis—Position that quantitative and qualitative research methods and philosophies can be combined

Complete counterbalancing—Enumerating all possible sequences and requiring different groups of participants to take different sequences

Concurrent probing—Obtaining a participant's perceptions of the experiment after completion of each trial

Concurrent validity—Degree to which scores obtained at one time correctly relate to the scores on a known criterion obtained at approximately the same time

Concurrent verbal report—A participant's oral report of the experiment, which is obtained as the experiment is being performed

Confidence interval—An interval estimate inferred from sample data that has a certain probability of including the true population parameter

Confidentiality—Not revealing identifying information obtained from a research participant to anyone outside the research group

Confounding—Occurs when an extraneous variable co-occurs with the independent variable and affects the dependent variable

Confounding extraneous variable—An extraneous variable that co-occurs with the independent variable and affects the dependent variable

Confounding variable—An extraneous variable that if not controlled for will eliminate the researcher's ability to claim that the IV causes changes in the DV

Constancy—The influence of an extraneous variable is same on all of the independent variable groups

Constant—Something that does not vary

Construct validity—The extent to which a construct is adequately represented by the measures used in a research study

Content-related evidence or content validity—Judgment by experts of the degree to which items, tasks, or questions on a test adequately represent the construct

Contingency question—An item directing participants to different follow-up questions depending on their response to the contingency question

Contingency table—Table used to examine the relationship between categorical variables

Control—(1) Manipulation of antecedent conditions to produce a change in mental processes and behavior, (2) eliminating the influence of extraneous variables, or (3) the group that does not receive the active ingredient believed to produce the effect

Control group—The group of participants that does not receive the active treatment condition and serves as a standard of comparison for determining whether the treatment condition produced any causal effect

Convenience sampling—Use of people who are readily available, volunteer, or are easily recruited for inclusion in a sample

Convergent validity evidence—Validity evidence based on the degree to which the focal test scores correlate with independent measures of the same construct

Correlation coefficient—Index indicating the strength and direction of linear relationship between two quantitative variables

Correlational research—A synonym for nonexperimental quantitative research

Counterbalancing—A technique used to control for sequencing effects

Counterfactual—What the experimental group participants' responses would have been if they had *not* received the treatment

Criterion-related validity—Degree to which scores predict or relate to a known criterion such as a future performance or an already-established test

Critical region—The area on a null hypothesis sampling distribution where the observed value of the statistic, if it fell in this area, would be considered a rare event

Cronbach's alpha—Another name for coefficient alpha

Cross-case analysis—Case study analysis in which cases are compared and contrasted

Cross-sectional design—Data are collected at a single time period

Crossover effect—An outcome in which the control group performs better at pretesting but the experimental group performs better at posttesting

Culture—The shared beliefs, values, practices, language, norms, rituals, and material things that the members of a group use to interpret and understand their world

Curvilinear regression—The type of regression analysis that can accurately model curved relationships

Curvilinear relationship—A nonlinear (curved) relationship between two quantitative variables

Cyclical variation—Any type of systematic up-and-down movement on the dependent variable over time

D

Data set—A set of data, where the rows are "cases" and the columns are "variables"

Data triangulation—Use of multiple sources of data

Debriefing—A postexperimental discussion or interview about the details of the study, including an explanation for the use of any deception

Deduction—A reasoning process that involves going from the general to the specific

Degrees of freedom—The number of values that are "free to vary"; it's used when computing a statistic to be used in inferential statistics

Dehoaxing—Debriefing the participants about any deception that was used in the experiment

Demand characteristics—Any of the cues available in an experiment, such as instructions, rumors, or setting characteristics, that influence the responses of participants

Dependent variable—Variable that is presumed to be influenced by one or more independent variables

Description—The portrayal of a situation or phenomenon

Descriptive NQR—Research focusing on providing an accurate picture or description of the status or characteristics of a situation or phenomenon

Descriptive statistics—The type of statistical analysis focused on describing, summarizing, or explaining a set of data

Descriptive validity—The factual accuracy of the account reported by the researcher

Desensitizing—Eliminating any undesirable influence that the experiment might have had on the participant

Design components—Structures and procedures used in constructing research designs

Determinism—The belief that mental processes and behaviors are fully caused by prior natural factors

Differential attrition—In a multiple-group design, groups become different on an extraneous variable because of differences in the loss of participants across the groups

Differential carryover effect—A treatment condition affects participants' performance in a later condition one way but in a different way when followed by a different condition

Differential history—The groups in a multiple-group design experience different history events that result in differences on the dependent variable

Differential influence—When the influence of an extraneous variable is different for the various groups

Direct effect—An effect of one variable directly on another variable; it is depicted as a single arrow in a path model

Directional alternative hypothesis—An alternative hypothesis that includes a "less than sign" (<) or a "greater than sign" (>)

Discoverability—The assumption that it is possible to discover the regularities that exist in nature

Discriminant validity evidence—Validity evidence based on the degree to which the focal test scores do *not* correlate with measures of *different* constructs

Disproportional stratified sampling—Stratified sampling where the sample proportions are made to be *different* from the population proportions on the stratification variable

Documents—Personal and official documents that were left behind

Double negative—A sentence construction that contains two negatives

Double-barreled question—Asking about two or more issues in a single question

Double-blind technique—Neither the researcher nor the participant is aware of the treatment condition administered to the participant

Duhem–Quine principle—States that a hypothesis cannot be tested in isolation from other assumptions

E

Ecological validity—The degree to which the results of a study can be generalized across settings or environmental conditions

Effect—The difference between what would have happened and what did happen when a treatment is administered

Effect size—The magnitude of the relationship between two variables in a population

Effect size indicator—Index of magnitude or strength of a relationship or difference between means

Element—The basic unit selected

Emic perspective—The insider's perspective

Emic–etic validity—Present when the researcher provides both the insider and objective outsider perspectives

Empirical—Research that is based on data

Empiricism—The acquisition of knowledge through experience

Equal probability of selection method (EPSEM)—Sampling method in which each individual element in the population has an equal probability of selection into the sample

Equal-status or integrative MMR—Study where both quantitative and qualitative data and approaches are of primary importance to the research

Equating the groups—Using control strategies to make the influence of extraneous variables constant across the independent variable groups so that the only systematic *difference* between the groups is due to the influence of the independent variable

Equivalent-forms reliability—Consistency of a group of individuals' scores on two versions of the same test

Essence—Phenomenological structure of the experience

Estimation—The branch of inferential statistics focused on obtaining estimates of the values of population parameters

Eta squared—The amount of variance in the dependent variable uniquely explained by a single independent variable

Ethical dilemma—The investigator's conflict in weighing the potential cost to the participant against the potential gain to be accrued from the research project

Ethnocentric—Judgment of people in other cultures based on the standards of your culture

Ethnography—Qualitative research method that focuses on the discovery and description of the culture of a group of people

Etic perspective—The researcher's external or "objective outsider" perspective

Event sampling—Observations are recorded every time a particular event occurs

Exhaustive categories—Response categories that cover the full range of possible responses

Existing or secondary data—Collection of data that were left behind or originally used for something different than the current research study

Experimental criterion—In single-case research, repeated demonstration that a behavioral change occurs when the treatment is introduced

Experimental group—The group of participants that receives the treatment condition that is intended to produce an effect

Experimental research—Research based on manipulation to demonstrate cause-and-effect relationships

Experimental-group-higher-than-control-group-at-pretest effect—An outcome in which the experimental group performs better than the control group at pretesting, and only the experimental group's scores change from pre- to posttesting

Experimental-group-lower-than-control-group-at-pretest effect—An outcome in which the control group performs better than the experimental group at pretesting, but only the experimental group improves from pre- to posttesting

Explanation—Knowledge of how and why a phenomenon operates, including its causes and the conditions under which the causes operate

Explanatory NQR—Nonexperimental research focused on how and why a phenomenon operates and in gaining some evidence of cause and effect

Extended fieldwork—Spending enough time in the field to fully understand what is being studied

External validity—Degree to which the study results can be generalized to and across other people, settings, treatments, outcomes, and times

Extraneous variable—Variable that might compete with the IV in explaining the outcome

F

Face validity—Prima facie judgment of whether the items appear to represent the construct and whether the test or instrument looks valid

Face-to-face interview method—Survey method where participants are interviewed in a face-to-face setting

Factor analysis—A statistical analysis procedure used to determine the number of dimensions present in a set of items

Factorial design—Two or more independent variables are studied to determine their separate and joint effects on the dependent variable

Factorial design based on a mixed model—A factorial design that uses a combination of within-participants and between-participants independent variables

Falsificationism—A deductive approach to science that focuses on falsifying hypotheses as the key criterion of science

Field experiment—An experiment that is conducted in a real-life or natural setting

Fieldnotes—Notes taken by the researcher during (or immediately after) observations in the field

Fieldwork—A general term for data collection in ethnographic research

Floor effect—Situation where participants' pretest scores on the dependent variable are too low to allow for additional decreases

Focus group—Collection of data in a group situation where a moderator leads a discussion with a small group of people

Frequency distribution—Data arrangement in which the frequencies of each unique data value is shown

G

Gatekeepers—Group members who control a researcher's access to the group

Generalizing validity—A synonym for external validity

Going native—Overidentification with the group being studied so that one loses any possibility of objectivity

Grounded theory—Methodology for generating and developing a theory that is grounded in the particular data

Group-administered questionnaire method—Survey method where participants fill out the questionnaire in a group setting

H

Histogram—Graph depicting frequencies and distribution of a quantitative variable

History effect—Any event that can produce the outcome, other than the treatment condition, that occurs during the study before posttest measurement

Holding the extraneous variable constant—The inclusion of participants from only one category or level of the extraneous variables

Holism—Idea that a whole, such as a culture, is more than the sum of its individual parts

Homogeneity—The degree to which a set of items measures a single construct

Hypothesis—The researcher's prediction or a tentative answer to a research question

Hypothesis testing—The process of testing a predicted relationship or hypothesis by making observations and then comparing the observed facts with the hypothesis or predicted relationship; the branch of inferential statistics focused on determining when the null hypothesis can or cannot be rejected in favor of the alternative hypothesis

I

Idiographic (or local) causation—An action for a particular person in a local situation with an observable result

Incomplete counterbalancing—Systematically enumerating fewer than all possible sequences and requiring different groups of participants to take the different sequences

Increasing-control-and-experimental-groups effect—An outcome in which the experimental and control groups differ at pretesting and both increase from pretest to posttest, but the experimental group increases at a faster rate

Independent samples *t* test—The significance test of the difference between two means that uses the *t* probability distribution

Independent variable—Variable that is presumed to cause changes in another variable

Indirect effect—An effect occurring through a mediating variable

Individual matching—A matching technique in which each participant is matched with another participant on selected variables

Induction—A reasoning process that involves going from the specific to the general

Inferential statistics—The type of statistical analysis focused on making inferences about populations based on sample data

Informed consent—Informing the research participant of all aspects of the study that might influence his or her willingness to volunteer to participate

Instrumental case study—Case study in which the researcher studies a case in order to understand something more general than the particular case

Instrumentation effect—Changes from pretest to posttest in the assessment or measurement of the dependent variable

Interaction effect—When the effect of two or more IVson the DV is more complex than indicated by the main effects

Interdependence—Violation of design assumption in which changing one target (participant, outcome, or setting) produces changes in the remaining targets

Internal consistency reliability—Consistency with which items on a test measure a single construct

Internal validity—The correctness of inferences made by researchers about cause and effect

Internet experiment—An experiment that is conducted over the Internet

Interobserver agreement—The percentage of time that different observers' ratings are in agreement

Interpretive (or emic) validity—Accurately portraying the participants' subjective viewpoints and meanings

Interrater reliability—The degree of consistency or agreement between two or more scorers, judges, observers, or raters

Interrupted time-series design—A quasi-experimental design in which a treatment effect is assessed by comparing the pattern of multiple pre- and posttest scores for a single group of research participants

Interval estimation—Placement of a range of numbers around a point estimate

Interval scale—A scale of measurement with equal intervals or distance between adjacent numbers

Interview—Verbal self-report data are collected from interviewees by an interviewer

Interview protocol—Data collection instrument used by the interviewer

Intrinsic case study—Case study in which the researcher is only interested in understanding the individual case

Investigator triangulation—Use of multiple investigators to collect and interpret the data

K

Known groups validity evidence—Degree to which groups that are known to differ on a construct actually differ *according to the test* used to measure the construct

L

Laboratory experiment—An experiment that is conducted in the controlled environment of a laboratory

Laboratory observation—Observation conducted in lab setting set up by the researcher

Leading question—A question that suggests how the participants should answer

Level of significance—Another name for alpha level

Lifeworld—A person's subjective inner world of experience

Likert scaling—A multi-item scale is used to measure a single construct by summing each participant's responses to the items on the scale

Line graph—A graph relying on the drawing of one or more lines connecting data points

Loaded term—A word that produces an emotionally charged reaction

Logic of discovery—The inductive or discovery part of the scientific process

Logic of hypothesis testing—The six steps in the process of significance testing

Logic of justification—The deductive or theory-testing part of the scientific process

Longitudinal cross-lagged design—A design in which two variables are measured at two or more points in time to determine the direction of relationship

Longitudinal design—Data are collected at two or more points in time

Low-inference descriptors—Descriptions that are very close to participants' words or are direct verbatim quotes

M

Mail questionnaire method—Survey method where questionnaires are sent to potential participants via regular mail

Main effect—The separate influence of one independent variable on the dependent variable

Manipulation—Researcher actively intervenes in the world by introducing a causal condition

Marginal mean—The average score of all participants receiving one level of an independent variable

Matching—Using any of a variety of techniques for equating participants on one or more variables

Matching variable—The extraneous variable used in matching; a variable a researcher ensures does not vary across levels of the IV to rule the matching variable out as a competing alternative explanation

Maturation—Any physical or mental change that occurs with the passage of time and affects dependent variable scores

Mean—The arithmetic average

Measure of central tendency—Numerical value expressing what is typical of the values of a quantitative variable

Measure of variability—Numerical value expressing how spread out or how much variation is present in the values of a quantitative variable

Measurement—The assignment of symbols or numbers to something according to a set of rules

Measurement research—Research focuses on determining the psychometric properties (reliability, validity, dimensionality, and operation across different kinds of people) of measurement instruments

Median—The center point in an ordered set of numbers

Mediating variable—Variable that occurs between two other variables in a causal chain; it's an intervening variable

Member checking—A synonym for participant feedback

Meta-analysis—Quantitative technique for describing the relationship between variables across multiple research studies

Method of data collection—Technique for physically obtaining the data to be analyzed in a research study

Method of difference—If groups are equivalent on every variable except for one, then that one variable is the cause of the difference between the groups

Methods triangulation—Use of multiple research methods or methods of data collection

Mixed methods research—Research in which quantitative and qualitative data or approaches are combined in a single study or set of related studies

Mixed sampling—Use of a combination of quantitative and qualitative sampling methods

Mixed-question format—Includes a mixture of both closed- and open-ended response characteristics in a single item

Mode—The most frequently occurring number

Moderator variable—Variable that changes or "moderates" the relationship between other variables

Multidimensional construct—Construct consisting of two or more dimensions; contrasted with a unidimensional construct

Multiple regression—Regression analysis with one dependent variable and two or more independent variables

Multiple validities legitimation—Making sure your mixed methods study meets appropriate quantitative, qualitative, and mixed methods validity types

Multiple-baseline design—A single-case design in which the treatment condition is successively administered to several target participants, target outcomes, or target settings

Multiple-stakeholder validity—Degree to which the study addresses interests, values, and viewpoints of different groups with a stake in the research

Mutually exclusive categories—Nonoverlapping response categories

N

Naturalistic generalization—Generalization, based on similarity, made by the reader of a research report

Naturalistic observation—Observation conducted in real-world situations

Negative correlation—Correlation in which values of two variables tend to move in opposite directions

Negative-case sampling—Searching for cases that challenge one's expectations or one's current findings

Nominal scale—The use of symbols, such as words or numbers, to classify or categorize measurement objects into groups or types

Nomological causation—The standard view of causation in science; refers to causal relationships among variables

Nondirectional alternative hypothesis—An alternative hypothesis that includes the "not equal to sign" (≠)

Nonequivalent comparison group design—A quasi-experimental design in which nonequivalent experimental treatment and comparison groups are tested at pretest and posttest

Nonexperimental quantitative research (NQR)—Type of quantitative research in which the independent or predictor variable is not manipulated by the researcher

Nonexperimental research—Research that lacks manipulation and studies the world as it naturally occurs

Nonmaleficence—Doing no harm to others

Nonnumerical data—Data that consist of pictures, words, statements, clothing, written records or documents, or a description of situations and behavior

Nonparametric statistics—Procedures that make fewer (or sometimes no) assumptions about populations and allow the analysis of nominal or ordinal data

Normal distribution—A theoretical distribution that follows the 68, 95, 99.7 percent rule

Norm group—The reference group upon which reported reliability and validity evidence is based

Norms—Unwritten and written rules specifying how people in a group are supposed to think and act

Null hypothesis—Typically, the hypothesis of no difference between means or no relationship in the population

Numerical data—Data consisting of numbers

O

Objectivity—Goal in science to eliminate or minimize opinion or bias in the conduct of research

Observation—Researcher watches and records events or behavioral patterns of people

One-group posttest-only design—Administration of a posttest to a single group of participants after they have been given an experimental treatment condition

One-group pretest–posttest design—Design in which participants are pretested, receive a treatment condition, and then are posttested on the dependent variable

One-stage cluster sampling—Clusters are randomly selected and all the elements in the selected clusters constitute the sample

One-way analysis of variance—Statistical test used when you have one quantitative DV and one categorical IV

One-way repeated measures analysis of variance—Statistical test used when you have one quantitative DV and one repeated measures IV

Open coding—First stage of data analysis in GT; it's the most exploratory stage

Open-ended question—A question that requires participants to respond in their own words

Operational definition—The result of operationalizing a construct

Operationalization—The way a construct is represented and measured in a particular research study

Operationalizing a construct—Identifying an instrument (or set of instruments) to accurately measure a construct

Order effect—A sequencing effect arising from the order in which the treatment conditions are administered to participants

Ordinal scale—A rank-order measurement scale

Outcome validity—The degree to which the results of a study can be generalized across different but related dependent variables

P

***p* value—**A shorter name for probability value in significance testing

Page number—The page number should appear in the upper right-hand corner of all manuscript pages. All pages should be numbered consecutively, beginning with the title page.

Panel studies—Longitudinal study where data are collected from the same individuals at successive time points

Paradigm emphasis—One of the two dimensions used in MM design matrix; its levels are equal status and dominant status

Parameter—A numerical characteristic of a population

Parametric statistics—Statistical analysis procedures that require many population assumptions

Partial correlation coefficient—The correlation between two quantitative variables controlling for one or more variables

Partial regression coefficient—The regression coefficient in a multiple regression equation

Partial-blind technique—A method whereby knowledge of each research participant's treatment condition is kept from the researcher and/or participant through as many stages of the experiment as possible

Participant feedback—Member checking to see if participants agree with the researcher's statements, interpretations, and conclusions

Participant observation—Data collection method in which the researcher becomes an active participant in the group being investigated

Participant reactivity to the research situation—Research participants' motives and tendencies that affect their perceptions of the situation and their responses on the dependent variable

Passive consent—Consent is assumed from a parent or guardian by not returning the consent form

Passive deception—Withholding information from the research participants by not giving them all the details of the experiment

Path analysis—Type of research in which a researcher hypothesizes a theoretical model and then tests the model with empirical data

Pattern matching—Construction and testing of a complex hypothesis

Peer review—Discussing your interpretations with peers and colleagues

Periodicity—Problematic situation in systematic sampling that can occur if there is a cyclical pattern in the sampling frame

Phenomenology—Qualitative research method where the researcher attempts to understand and describe how one or more participants experience a phenomenon

Physical data—Any material thing created or left behind by humans that might provide clues to some event or phenomenon

Pilot study—A preliminary study that is conducted on a few participants prior to the actual research study

Pilot test—Testing for proper operation of a data collection instrument before using it in the research study

Placebo effect—Improvement due to participants' expectations for improvement rather than the actual treatment

Plagiarism—Using work produced by someone else and calling it one's own

Point estimation—Use of the value of a sample statistic as one's estimate of the value of a population parameter

Population—The full group of interest to the researcher that the researcher wants to generalize to, and from which the sample is selected.

Population validity—Degree to which the study results can be generalized to and across the people in the target population

Positive correlation—Correlation in which values of two variables tend to move in the same direction

Positive self-presentation—Participants' motivation to respond in such a way as to present themselves in the most positive manner

Post hoc tests—Follow-up test to ANOVA when the categorical IV has three or more levels; used to determine which pairs of means are significantly different

Postexperimental inquiry—Asking the participants for their impressions of the study after they complete the study

Posttest-only control-group design—Administration of a posttest to two or more randomly assigned groups of participants that receive the different levels of the independent variable

Posttest-only design with nonequivalent groups—Design in which the performance of an experimental group is compared with that of a nonequivalent control group at the posttest

Power—The probability of rejecting a false null hypothesis

Practical significance—Claim made when a statistically significant finding seems large enough to be important

Pragmatic validity—Extent to which the research purpose(s) was met, research questions adequately answered, and actionable results provided.

Pragmatism—Philosophy focusing on what works as the criterion of what should be viewed as tentatively true and useful in research and practice

Prediction—The ability to anticipate the occurrence of an event

Predictive NQR—The use of one or more predictor variables to make predictions about outcomes occurring in the future

Predictive validity—Degree to which scores obtained at one time correctly predict the scores on a criterion at a later time

Pretest–posttest control-group design—Administration of a posttest to two or more randomly assigned groups of participants after the groups have been pretested and administered the different levels of the independent variable

Privacy—Controlling other people's access to information about a person

Probabilistic causes—A weaker form of determinism that indicates regularities that usually but not always occur

Probability value—The likelihood of the observed value (or a more extreme value) of a statistic, if the null hypothesis were true

Propensity score matching—A statistical procedure used in nonexperimental quantitative research that equates the independent variable groups on multiple extraneous variables

Proportional stratified sampling—Stratified sampling where the sample proportions are made to be the *same* as the population proportions on the stratification variable

Proximal similarity—Generalization to people, places, settings, and contexts that are similar to those described in the research study

Pseudoscience—Practices, beliefs, and procedures that claim to be scientific but are not.

Psychological experiment—Establishes the presence or absence of a relationship by manipulating an independent variable under controlled conditions and measuring the effect on a dependent variable.

PsycINFO—An electronic bibliographic database of abstracts and citations to the scholarly literature in psychology

Purpose of random assignment—To produce two or more equivalent groups for use in an experiment

Purpose of random selection—To obtain a representative sample

Purposive sampling—A researcher specifies the characteristics of the population of interest and then locates individuals who have those characteristics

Q

Qualitative research—Research based on nonnumerical data

Qualitatively driven MMR—Study where supplemental quantitative data are added to a generally qualitative research study

Quantitative research—Empirical research that relies on quantitative data and approaches

Quantitative variable—Variable that varies by degree or amount

Quantitatively driven MMR—Study where supplemental qualitative data are added to a generally quantitative research study

Quasi-experimental design—A research design in which an experimental procedure is applied but all extraneous variables are not controlled

Questionnaire—Self-report data collection instrument filled out by research participants

Quota sampling—A researcher decides on the desired sample sizes or quotas for groups identified for inclusion in the sample, followed by convenience sampling from the groups

R

Random assignment—Control technique that equates groups of participants by ensuring every member an equal chance of being assigned to any group

Random sampling—Selection of sample members using a statistically random process

Random selection—Selection of participants using a random sampling method

Random-digit dialing—Random sampling method frequently used with telephone interviewing

Randomization—A synonym for random assignment. It refers to the chance process

Randomized controlled trial (RCT)—Experimental design with random assignment to experimental (treatment) and control groups

Randomized counterbalancing—Sequence order of the treatment conditions is randomly determined for each individual

Randomized designs—Between-participants designs in which participants are randomly assigned to groups

Range—The highest number minus the lowest number

Ranking—Participants asked to put their responses in ascending or descending order

Rates—The percentage of people in a group that have a particular characteristic

Rating scale—An ordered set of response choices, such as a 5-point rating scale, measuring the direction and strength of an attitude

Ratio scale—A scale of measurement with rank ordering, equal intervals, and an absolute zero point

Rationalism—The acquisition of knowledge through reasoning

Reactive effect—Nontypical behavior of participants because of the presence of the researcher

Reality in nature—The assumption that the things we see, hear, feel, smell, and taste are real

Reflexivity—Thinking critically about one's interpretations and biases

Regression analysis—Use of one or more quantitative independent variables to explain or predict the values of a single quantitative dependent variable

Regression artifacts—Effects that appear to be due to the treatment but are due to regression to the mean

Regression coefficient—The slope or change in Y given a one unit change in X

Regression discontinuity design—A design that assigns participants to groups based on their scores on an assignment variable and assesses the effect of a treatment by looking for a discontinuity in the groups' regression lines

Regression equation—The equation that defines a regression line

Regression line—The line of "best fit" based on a regression equation

Regression toward the mean—A synonym for regression artifacts

Reliability—The consistency or stability of scores

Reliability coefficient—Type of correlation coefficient used as an index of reliability

Repeated measures design—Another common name for a within-participants design

Replication—The reproduction of the results of a study in a new study

Representative sample—A sample that resembles the population

Research design—The outline, plan, or strategy used to investigate the research question

Research ethics—A set of guidelines to assist researchers in conducting ethical research

Research misconduct—Fabricating, falsifying, or plagiarizing the proposing, performing, reviewing, or reporting of research results

Research question—Any question that asks about the relationship between at least two variables

Research validity—Truthfulness of inferences made from a research study

Researcher attributes—Biasing experimenter effects attributable to the physical and psychological characteristics of the researcher

Researcher bias—Only noticing data that support one's prior expectations

Researcher expectancies—Biasing experimenter effects attributable to the researcher's expectations about the outcome of the experiment

Researcher or experimenter effects—Actions and characteristics of researchers that influence the responses of participants

Researcher-as-detective—Metaphor applied to researcher looking for the local cause of a single event

Response rate—The percentage of people selected to be in a sample who actually participate in the research study

Response set—Tendency for a participant to respond in a particular way to a set of items

Retrospective verbal report—An oral report in which the participant retrospectively recalls aspects of the experiment

Reversal—Change of behavior back to baseline level after withdrawal of treatment

Rival hypothesis—Another name or synonym for alternative explanation; refers to the possibility of extraneous variables explaining the relation of the IV to the DV

S

Sacrifice groups—Groups of participants who are stopped and interviewed at different stages of the experiment

Sample—A subset of the population; the set of cases or elements selected from the population

Sample integration validity—Researchers must not treat the quantitative and qualitative samples as equal, but, instead, draw appropriate conclusions from each sample

Sample size calculator—A statistical program used to provide the recommended sample size

Sampling—The process of drawing a sample from a population

Sampling distribution—The theoretical probability distribution of the values of a statistic that would result if you selected all possible samples of a particular size from a population

Sampling distribution of the mean—The theoretical probability distribution of the means of all possible samples of a particular size selected from a population

Sampling error—Differences between sample values and the true population parameter

Sampling frame—A list of all the elements in a population

Sampling interval—The population size divided by the desired sample size; it's symbolized by the letter *k*

Scatterplot—A graphical depiction of the relationship between two quantitative variables

Science—The approach to producing knowledge used by researchers employing the methods of science

Search engine—A software program that seeks out web pages stored on servers throughout the World Wide Web

Seasonal variation—Values on the dependent variable vary by season

Selection—Production of nonequivalent groups because a different selection procedure operates across the groups

Selection-attrition effect—Participants who drop out of one group are dissimilar to those in another group

Selection-history effect—An extraneous event occurring between pretest and posttest influences participants in one group differently than participants in another group

Selection-instrumentation effect—Participants' scores in one group are affected by changes in the process of measurement differently than participants in another group

Selection-maturation effect—Participants in one group experience a different rate of maturation than participants in another group

Selection-regression-artifact effect—Participants in one group display a different rate of regression to the mean than participants in another group

Selection-testing effect—Previous experience with the test affects the groups differently

Selective coding—Third and final stage of data analysis in GT in which the theory is finalized

Semantic differential—Scaling method measuring the meanings that participants give to attitudinal objects

Semi-partial correlation squared—The amount of variance in the dependent variable uniquely explained by a single quantitative independent variable

Sequential validity—Making sure that the ordering of quantitative and qualitative components in a sequential design does not bias the results, and making sure that later phases appropriately build on earlier phases in the design.

Shared beliefs—Statements or conventions that people sharing a culture hold to be true or false

Shared values—Culturally defined standards about what is good or bad or desirable or undesirable

Significant statements—Words, phrases, or sentence length participant statements that the researcher thinks vividly communicate the participant's experience

Simple case of NQR—Situation in NQR where there are only two variables and the researcher does not control for any extraneous variables

Simple random sampling—A popular and basic equal probability of selection method

Simple regression—Regression analysis with one dependent variable and one independent variable

Single-blind technique—A method where the research participant is not aware of the treatment condition he or she is in.

Single-case research design—Research design in which a single participant or a single group of individuals is used to investigate the influence of a treatment condition

68, 95, 99.7 percent rule—Rule stating percentage of cases falling within 1, 2, and 3 standard deviations from the mean on a normal distribution

Snowball sampling—Each sampled person is asked to identify other potential participants with the inclusion characteristic

Social comparison method—A social validation method in which the participant is compared with typical peers

Social desirability bias—Error occurring when participants try to respond in the way they think makes them look good

Social validation—Determination by others that the treatment condition has significantly changed the participant's functioning

Specificity of the research question—The preciseness with which the research question is stated

Spurious relationship—A relationship between two variables that is not causal

Stable baseline—A set of responses characterized by the absence of trend and little variability

Standard deviation—The square root of the variance

Standard error—The standard deviation of a sampling distribution

Statistic— A numerical characteristic of sample data

Statistical conclusion validity—The validity of inferences made about the covariation between the independent and dependent variables

Statistical control—Control of measured extraneous variables during data analysis

Statistical power—The probability of correctly rejecting the null hypothesis when it is false

Statistically significant—Conclusion that an observed finding would be very unlikely if the null hypothesis were true

Stratification variable—The variable on which the population elements are divided for the purpose of stratified sampling

Stratified random sampling—Division of population elements into mutually exclusive groups and then selection of a random sample from each group

Strong experimental designs—Designs that effectively control extraneous variables and provide strong evidence of cause and effect

Subject matching—Synonym for individual matching

Subjective evaluation method—Social validation method where others are asked if they perceive a change in the participant's behavior

Summated rating scale—Another name for Likert scaling

Survey instrument—Data collection instrument used in survey research such as a questionnaire or interview protocol

Survey research—A nonexperimental research method relying on questionnaires or interview protocols

Systematic sampling—The sampling method where one determines the sampling interval (k), randomly selects an element between 1 and k, and then selects every kth element

T

***t* test for correlation coefficient—**Statistical test used to determine if a correlation coefficient is statistically significant

***t* test for regression coefficients—**Statistical test used to determine if a regression coefficient is statistically significant

Target population—The large population to which the researcher would like to generalize the study results

Telephone interview method—Survey method where interviews are conducted over the telephone

Temporal validity—The degree to which the results can be generalized across time

Test statistic—A statistic that follows a known sampling distribution and is used in significance testing

Testing effect—Changes in a person's score on the second administration of a test resulting from having previously taken the test

Tests—Standardized or researcher-constructed data collection instruments designed to measure personality, aptitude, achievement, and performance

Test–retest reliability—Consistency of a group of individuals' scores on a test over time

Theoretical generalization—Generalization of a theoretical explanation beyond the particular research study

Theoretical sampling—Continual selection of new participants during a study based on who might be most relevant for developing a good theoretical explanation

Theoretical saturation—Occurs when no new information relevant to the GT is emerging from the data and the GT has been sufficiently validated

Theoretical sensitivity—Researcher is effective in understanding what kinds of data need to be collected and what aspects of already collected data are important for theory development

Theoretical validity—Degree to which the theory or explanation fits the data

Theory—An explanation of how and why something operates; a statement or group of statements that explain "how" and "why" some part of the world works and provides predictions about the world to be empirically tested

Theory triangulation—The use of multiple theories or perspectives to aid in interpreting the data

Therapeutic criterion—Demonstration that the treatment condition has improved everyday functioning in ways that are clinically or socially relevant

Think-aloud technique—A method that requires participants to verbalize their thoughts as they engage in an activity

Third variable—A synonym for a confounding extraneous variable

Third variable problem—Occurs when observed relationship between two variables is actually due to a confounding extraneous variable

Three-way interaction—A two-way interaction that changes at the different levels of the third independent variable

Time order—One of the two dimensions used in MM design matrix; its levels are concurrent and sequential

Time-interval sampling—Observations are recorded during preselected time intervals

Treatment variation validity—The degree to which the results of a study can be generalized across variations in the treatment

Trend study—Independent samples are taken successively from a population over time and the same questions are asked

Triangulation—Use of multiple data sources, research methods, investigators, and/or theories/perspectives to cross-check and corroborate research data and conclusions

Two-stage cluster sampling—Clusters are randomly selected, and a random sample of elements is drawn from each of the selected clusters

Two-way analysis of variance—Statistical test used when you have one quantitative DV and two categorical IVs

Two-way interaction—The effect of one independent variable on the dependent variable varies across the different levels of the other independent variable

Type I error—Rejection of a true null hypothesis

Type II error—Failure to reject a false null hypothesis

U

Unstandardized difference between means—The difference between two means in the variables' natural units

V

Validation—Gathering of evidence regarding the soundness of inferences made from test scores

Validity—Accuracy of inferences, interpretations, or actions made on the basis of test scores

Validity coefficient—The type of correlation coefficient used in validation research

Variable—A condition or characteristic that can take on different values or categories

Variance—The average deviation of data values from their mean in squared units

Volunteer sampling—Nonrandom sampling method where participants self-select into the sample

W

Weak experimental designs—Designs that do not control for many extraneous variables and provide weak evidence of cause and effect

Weakness minimization validity—Present when the researcher compensates for the weakness of one approach through the use of an additional approach

Web-based survey—Electronic survey where participants are contacted and complete the measure online

Within-participants design—All participants receive all conditions (uses a within-participants IV)

Within-participants independent variable—Type of independent variable where all participants receive all levels of the independent variable

Within-participants posttest-only design—All participants receive all conditions, and a posttest is administered after each condition is administered

Within-participants posttest-only design—only design All participants receive all conditions, and a posttest is administered after each condition is administered

Y

***Y*-intercept**—The point at which a regression line crosses the *Y* (vertical) axis

Z

***z* score**—A score that has been transformed into standard deviation units